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Thermodynamic equilibrium analysis of combined dry and steam reforming of propane for thermochemical waste-heat recuperation

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ABSTRACT

The thermochemical waste-heat recuperation is one for perspective way of increasing the energy efficiency of the fuel-consuming equipment. In this paper, the thermochemical waste-heat recuperation (TCR) by combined steam-dry propane reforming is described. To understand the influence of technological parameter such as temperature and composition of inlet gas mixture on TCR efficiency, thermodynamic equilibrium analysis of combined steam-dry propane reforming was investigated by Gibbs free energy minimization method upon a wide range of temperature (600–1200 K) and different feed compositions at atmospheric pressure. The carbon and methane formation was also calculated and shown. From a thermodynamic perspective, the TCR can be used for increasing energy efficiency at temperatures above 950 K because in this range the maximum conversion rate is reached (from 1.22 to 1.30 for the different feed composition). Approximately 10 mol of synthesis gas can be generated per mole of propane at the temperatures greater than 1000 K. Furthermore, the propane conversion rate and yield of hydrogen are increased with the addition of extra steam to the feed stock. Also, undesirable carbon formation can be eliminated by adding steam to the feed. The thermodynamic equilibrium analysis was accomplished by IVTANTHERMO which is a process simulator for thermodynamic modeling of complex chemically reacting systems and several results were checked by Aspen-HYSYS.

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Introduction

The experts of the International Energy Agency showed that fossil fuel such as, oil and natural gas, will be the primary fuel for the World Economy during the future 100–150 years [1]. Nearly 80% of energy demand in the world by 2040 will be supplied by fossil fuels and the share of natural gas will reach 26% of the global demand for energy from the current 21% in 2015. Gas fuel such as propane is very important commodities because it is the main feeds of different industrial sectors [2]: transport [3–7], gas turbine [8–10], agriculture [11–13], e.a. Propane has replaced many older other traditional fuel sources. Propane is clean burning and effectively competes with other fossil and renewable fuels on efficiency and greenhouse gas emissions in many applications. Its simple chemical make-up allows it to burn cleaner than coal, light and heavy petroleum fuels, ethanol, and even natural gas in some cases. It is expected that the usage of propane in the future will increase continuously. Therefore, problems of improving energy

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efficiency get more relevant for the many propane-consuming equipment.

A common way to improve the energy efficiency of such equipment is heating of air for combustion by means of cooling flue gases [14–18]. This leads to significant increase of energy efficiency [19], but that way of recovery has some disadvantages. The heat exchangers for the preheat air have a large size, because heat-transfer coefficient in such exchangers is very low [20,21]. Preheat of propane in the heat exchangers by means of flue gases is not widely used, because it leads to high temperature propane cracking [22,23] which results in a reduction of heat-transfer coefficient in the exchangers. These disadvantages can eliminated using thermochemical heat recuperation of waste flue gases [24–27]. Thermochemical recuperation (TCR) is a perspective way to increase energy efficiency of the much propane-consuming equipment, for example, industrial propane-consumed gas furnaces, gas turbines and others.

Thermochemical recuperation of waste flue gases occurs when chemical endothermic reactions are used, for example, steam methane reforming [26], combined steam-dry methane reforming [28], steam ethanol reforming [25], propane reforming, etc. The idea of improving the efficiency of an energy plant by using the chemical recuperation of waste heat was discussed as early as in 1972 [24]. In recent years, the questions of chemical recuperation have been discussed in many papers [25,27–30].

Generally, for the thermochemical heat recovery can be used different chemical endothermic reaction, but the most obvious its advantages and feasibility of using a reforming of hydrocarbon gas, such as methane and propane [26]. To understand the effects of process variables such as temperature and composition of inlet gas mixture on TCR efficiency, thermodynamic equilibrium analysis of combined steam-dry propane reforming was investigated by Gibbs free energy minimization method upon a wide range of temperature (600–1200 K) and different feed compositions at atmospheric pressure.

Investigate studies of steam and dry propane reforming [31–35] were generally concentrated on research of catalyst performance and the results of these researches cant be matched because its the big difference of the operation conditions (temperature and inlet reactant composition). Nevertheless, the influence of the feed composition and temperature on equilibrium conversion rate and outlet composition can be determined by thermodynamic analysis. Different catalysts have been studied in propane reforming, including nickel-based [36–39] and noble metal-based catalysts (Pt, Rh, Pd, etc.) [40–43]. Concerning dry reforming of propane, Raaberg et al. [36,37] showed an exceptionally stable Ni/Mg(Al)O hydrotalcite derived catalyst. The intrinsic activity of catalysts depended strongly on Ni particle size. Further mechanistic studies showed C–C bond rupture is the rate determining step. Solymosi et al. [42,44] analyzed the characteristic of catalysts based on noble metal such as platinum (Pt), rhodium (Rh) and ruthenium (Ru) and reported the highest specific rates for hydrogen and carbon monoxide yield were on catalysts based on rhodium and ruthenium. Researching of catalyst based on bimetallic composition Co–Ni/Al_2O_3 in both CO_2 and H_2O propane reforming, it was concluded that coke deposition is the major reason of catalyst deactivation and poisoning [39,45–47]. Wang et al. [48] reported the results of thermodynamic equilibrium calculations steam and dry propane reforming which were obtained by equilibrium software Outokumpu HSC Chemistry 4.0, using the extensive thermochemical database delivered in the software package. Zheng et al. [49] reported a thermodynamic analysis of propane OSR in fuel cell for hydrogen and carbon monoxide production. However, there are no articles or reports which are contained results of thermodynamic equilibrium analysis of combined steam-dry propane reforming in terms of further influence of inlet feed ration and temperature by means of total Gibbs free energy minimization of the process (according to the author).

In this paper, author report the thermodynamic analysis of combined steam-dry reforming of propane, where total Gibbs free energy minimization method was employed to calculate equilibrium compositions for thermochemical waste-heat recuperation and transformation rate which is used to describe the relation of low heat value (LHV) of new synthesis fuel to LHV of initial propane. The carbon and methane formation was also calculated and illustrated. The effect of pressure on combined steam-dry propane reforming is not considered due to schematic diagram that is shown bottom on Fig. 1.

Thermochemical recuperation concept

Schematic diagram

The main concept of thermochemical waste-heat recuperation is transformation of flue gases heat into chemical energy of a new synthetic fuel that has higher calorimetric properties such as low heating value [50]. If in the conventional fuel-consuming equipment, for example the industrial furnaces, conversion of chemical energy of fuel into heat energy occurs by fuel combustion, then in the industrial furnaces with TCR the conversion of chemical energy of fuel is divided into two stages. The first stage is the increasing of the low heat value of initial propane by transformation of enthalpy waste-heat into chemical energy of the new synthetic fuel. The second stage is combustion of the new synthetic fuel that has low heat value greater than the low heat value of propane.

Fig. 1 shows the schematic diagram of the propane-consuming equipment with thermochemical heat recuperation of waste flue gases by propane reforming with flue gases.

In this diagram, the flue gases after the combustion chamber of the propane-consuming equipment are divided into two streams. First stream is fed to a reaction volume of the reformer. Also, propane is fed to the reformer. The reaction space of the reformer is activated by nickel catalysts [37–39,51–53]. In the reactor volume of the reformer occur reactions of the steam and dry reforming of propane and other chemical reactions which produce synthesis gas [54]. The produced gas (new synthetic fuel) is fed to the combustion chamber for combustion [55–59]. Second stream from the combustion chamber sequentially is passed through a heat-exchange space of the reformer and the air heater. The cool flue gases after the air heater are ejected into atmosphere. Second stream is passed through the heat-exchange space of

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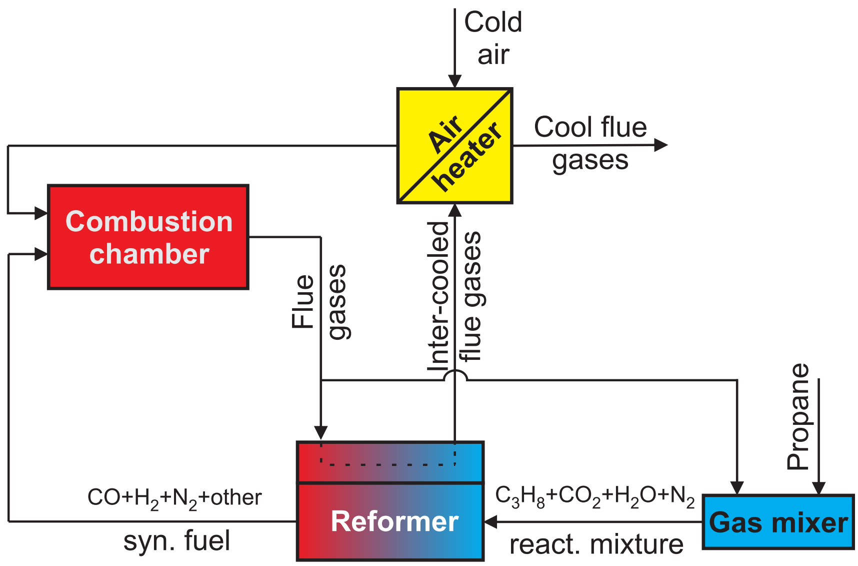

Fig. 1 – The schematic diagram of propane-consuming equipment with thermochemical waste-heat recuperation.

the reformer because the reactions of steam and dry propane reforming are endothermic and its completion requires external heat supply. The stream of inter-cooled flue gases is passed to the air heater for preheating of cold air.

According Fig. 1 the composition of flue gas, reaction mixture and synthesis gas depends on different conditions. To determine the composition of the synthesis gas, depending on the operation conditions, the thermodynamic analysis was carried out. Table 1 presents gas compositions which determined below as $mol_i/mol_{mix}\%$ at every point for air-fuel equivalence ratio of unit $\lambda = 1$ at combustion chamber ($C_3H_8 + 5\ (O_2 + 3.76N_2)$).

Chemical reactions

The overall reaction in steam-dry reforming process to form H_2, C, CO, CO_2, H_2O and other components may include the equations listed in Table 2 [48]. Nitrogen that contains in the flue gases is chemically neutral, because the contribution of reactions involving nitrogen in the energy and material balance of the overall reforming process is negligible. Other chemical reactions may also occur in the reformer, but contribution of those reactions for energy and material balance of overall reforming process is negligible [60–62].

Obviously, the efficiency of use of thermochemical waste-heat recovery significantly depends on the conversion rate of propane (reactions 1–2) because it indicates how much of waste-heat of flue gases was transformed into chemical energy of new synthetic fuel. Thermodynamic equilibrium analysis of combined steam-dry propane reforming allows to determine influence of feed composition and temperature on conversion rate of propane.

Table 1 – Gas composition as mol_i/mol_{mix}, % at every flow in Fig. 1

Point	Gas composition	mol/mol, %
Hot air	O_2/N_2	21/79
Flue gases	$N_2/H_2O/CO_2$	72.9/15.5/11.6
Reaction mixture	$C_3H_8/H_2O/CO_2/N_2$	3.7/14.9/11.2/70.2
Synthesis gas	H_2:CO:H_2O:CO_2:N_2:other	
	T = 1200 K	18.4/12.3/6.6/5.9/56.8/0.0
	T = 900 K	8.6/5.9/6.2/7.8/63.4/8.1

Methodology

The method of thermodynamic analysis of various chemical process by minimization of Gibbs free energy was introduced in the previous publications [25] and it was also described in detail by other authors [49,63–66]. Thermodynamic analysis of combined steam-dry reforming of propane via total Gibbs free energy minimization method was introduced to determine equilibrium compositions. The method of Gibbs free energy minimization is based on the

Table 2 – Possible chemical reactions in the reformer [35,48].

No.	Reaction	ΔH_{298}, (kJ/mol)
The overall reaction of SR and DR		
1	$C_3H_8 + 6H_2O \Leftrightarrow 3CO_2 + 10H_2$	374.1
2	$C_3H_8 + 3CO_2 \Leftrightarrow 6CO + 4H_2$	620.3
Water gas shift reaction		
3	$CO + H_2O \Leftrightarrow CO_2 + H_2$	−41.1
Methanation		
4	$CO + 3H_2 \Leftrightarrow CH_4 + H_2O$	−206.1
5	$CO_2 + 4H_2 \Leftrightarrow CH_4 + 2H_2O$	−165.0
Dry propane reforming		
6	$CH_4 + CO_2 \Leftrightarrow 2H_2 + 2CO$	247.3
Carbon formation [43]		
7	$C_3H_8 \Leftrightarrow C_2H_4 + CH_4$	89.0
8	$C_2H_4 \Leftrightarrow 2C + 2H_2$	−52.2
9	$H_2 + CO \Leftrightarrow C + H_2O$	−131.1
10	$CH_4 \Leftrightarrow C + 2H_2$	74.8
11	$2CO \Leftrightarrow C + CO_2$	−173.3
12	$CO_2 + 2H_2 \Leftrightarrow C + 2H_2O$	−90.13

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principles that the analyzed chemical scheme is thermodynamically auspicious if total Gibbs free energy has lowest value and differential of Gibbs free energy is zero for a presented operation conditions [67]:

$$dG^t = 0 \tag{1}$$

where G^t is total Gibbs free energy.

At low pressure and high temperature, the system can be considered as ideal [68]. Moreover, the reaction products from the reformer are determined, based on the following admissions: steady flow process with negligible change of kinetic and potential energies; perfect insulation of reformer with negligible heat loss; chemical reactions with known reactants and products compositions are equilibrium.

Total Gibbs free energy of a chemical system is presented by the sum of the chemical potential of all chemical elements:

$$G^t = \sum n_i \mu_i \tag{2}$$

where n_i is number of moles of species i; μ_i is chemical potential of species i.

The chemical potential of each element is determined as follow:

$$\mu_i = \Delta G_{fi}^0 + RT \ln\left(\frac{f_i}{f_i^0}\right) \tag{3}$$

where G_{fi}^0 is standard Gibbs free energy of formation of species i; R is molar gas constant; T is temperature of the system; f_i is fugacity of pure species i; f_i^0 is fugacity of pure species i at the standard state.

If combine the expressions above, the equation for total Gibbs free energy of the chemical system can be written as:

$$G^t = \sum n_i \Delta G_{fi}^0 = \sum n_i RT \ln\left(\frac{f_i}{f_i^0}\right) \tag{4}$$

The thermodynamic analysis of considered chemical system introducing the Gibbs free energy minimization technique was accomplished by IVTANTHERMO which is a process simulator for thermodynamic modeling of complex chemically reacting systems [69–71].Software IVTANTHERMO contains an extensive database on thermodynamic properties of chemical components, algorithms for the database computing and a software, which allows the determination of equilibrium composition and thermodynamic properties of the chemical system to be examined. The primary benefit of this software is its opportunity to find reliably the phase composition of the system when possible phase number is large. In present investigation this is important for calculation of carbone formation. In case of propane reforming the number of phases is at least two. Besides several results were checked by Aspen-HYSYS version 8.4, which is a powerful software package for modeling unsteady-state and steady-state chemical process for different systems [72–75]. Also, both programs calculate the distribution of reaction products by minimizing the Gibbs free energy of each of the existing species in the reaction system regardless of the reaction network.

For all the results obtained below the difference between the data obtained with the help of IVTANTHERMO and Aspen-HYSYS was less than 1% for all cases. Therefore, the graphs below show data obtained only with the help of IVTANTHERMO for better understanding of graphs.

Carbon formation in gas phase reactions can be calculated by exploiting the phase equilibrium existed between solid carbon and vapor carbon in the gas phase. Considering all carbon to be in graphite form with the free energy of formation as zero and to have no vapor pressure in the range of temperatures studied in this work, then the total Gibbs free energy can be considered to be independent of carbon [43,48,76].

The major gas species involved in combined steam-dry reforming of propane are propane (C_3H_8), water (H_2O), hydrogen (H_2), carbon monoxide (CO), carbon dioxide (CO_2), methane (CH_4), acetylene (C_3H_4), ethylene (C_3H_6) and carbon (graphite) based on experimental results [32,48]. The most occurring species, including H_2, CO, CO_2, CH_4 and carbon [43,77], together with C_3H_8 and H_2O were considered in this work. The dominant steam reforming and dry reforming reactions are strongly endothermic.

The equilibrium conversion rates of reactants and the H_2 and CO yield are determined as:

$$X_{C_3H_8}(\%) = \frac{[C_3H_8]_{in} - [C_3H_8]_{out}}{[C_3H_8]_{in}} \times 100\% \tag{5}$$

$$X_{CO_2}(\%) = \frac{[CO_2]_{in} - [CO_2]_{out}}{[CO_2]_{in}} \times 100\% \tag{6}$$

$$X_{H_2O}(\%) = \frac{[H_2O]_{in} - [H_2O]_{out}}{[H_2O]_{in}} \times 100\% \tag{7}$$

$$Y_{CO}(\%) = \frac{[CO]_{out}}{[CO_2]_{in} + 3[C_3H_8]_{in}} \times 100\% \tag{8}$$

$$Y_{H_2}(\%) = \frac{2[H_2]_{out}}{8[C_3H_8]_{in} + 2[H_2O]_{in}} \times 100\% \tag{9}$$

Transformation rate (η_q) is used to describe the relation of low heat value of new synthesis fuel to LHV of initial propane:

$$\eta_q = \frac{LHV_{synfuel}}{LHV_{C_3H_8}} \tag{10}$$

Low heat value of new synthesis fuel is given as sum of LHV of combustible components of synfuel after reformer (according schematic diagram on Fig. 1):

$$LHV_{synfuel} = \sum n_{mol} \cdot LHV_{C_xH_y} \tag{11}$$

The initial value of propane is assumed to be 1 mol. First of all, equilibrium changes of the conversion rates and product yields were investigated as a function of temperature (600–1200 K) at different composition of inlet mixture (Table 3) for constant atmospheric pressures 1 bar.

The feed compositions (a) and (b) are chosen to compare the thermodynamic results with other studies [48]. The feed composition (c) corresponds to the composition of the combustion products of propane [78,79]. In the schematic diagram shown in Fig. 1, the feed composition can be varied by the addition of steam into the gas mixer. Therefore, the different feed compositions (d–f) are used. The feed composition (g) shows twice more the combustion products into gas mixer.

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Table 3 – Composition of inlet gas mixture.

Feed composition	Oxidant to propane ratio	Moles of H_2O	Moles of CO_2
a	3	3	0
b	3	0	3
c	7	4	3
d	8	5	3
e	9	6	3
f	12	9	3
g	14	8	6

Results and discussion

Conversion rate

Complete propane conversion was received for the range of temperature and feed composition at atmospheric pressure. Fig. 2 shows equilibrium changes of the propane conversions as a function of temperature when different feed composition is introduced into the reformer (according Fig. 1) at atmospheric pressure.

Increase moles of steam (H_2O) in feed stream have a notable influence on propane conversion rate. The addition of supplementary H_2O into the feed stock increased $X_{C_3H_8}$ for few percent for each feed composition; primarily at temperatures to 1000 K. This fact can be explained because the addition of H_2O increased propane activity in overall combined steam-dry reforming process. The maximum conversion rate (close to 100%) is observed for all feed compositions where are H_2O and CO_2 (Table 3) Above 1000 K, steam additions has no effects on $X_{C_3H_8}$ because propane becomes the limiting reacting component and is consumed in both steam and dry reactions according Table 2. The obtained results for the H_2O and CO_2 reforming of propane (a and b feed compositions in Table 3) are well correlated with the results of other researchers [48].

Fig. 3 illustrates that although higher numbers of steam moles resulted in more propane conversion rate, increasing

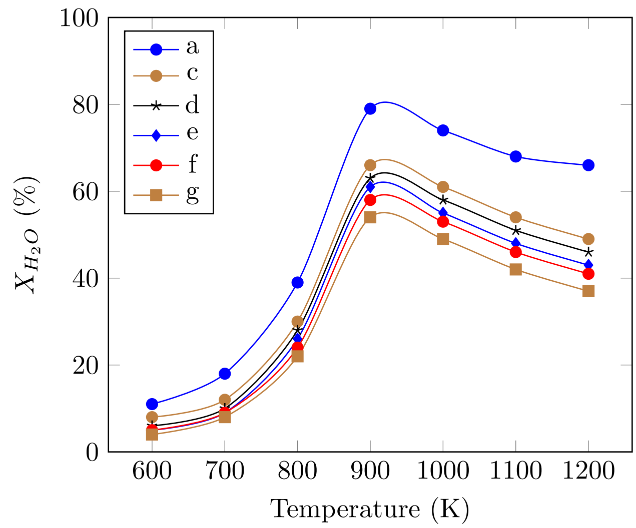

Fig. 3 – Equilibrium conversion rate of steam (H_2O).

H_2O content didn't favor for steam conversion rate considering that the fixed number of moles of propane and carbon dioxide in the feed stock.

The conversion rate of steam increased in temperature range until 910 K where the equilibrium conversion reached its maximum level for all feed composition. It happens due to the fact that until 910 K the prevalent reactions are water-gas shift and steam propane reforming. The steam conversion rate decreased at temperature range above 920 K due to the fact that reverse water-gas shift reaction took place, which produced extra steam. Also at than temperature range dry propane reforming occurred due to its more endothermic nature than steam propane reforming. Moreover, the maximum rate of steam conversion shifted to left for higher numbers of steam moles in the feed stock.

Fig. 4 shows the conversion rate of carbon dioxide (CO_2) as a function of temperature at the different feed stock for atmospheric pressure. CO_2 conversion rate decreased with an

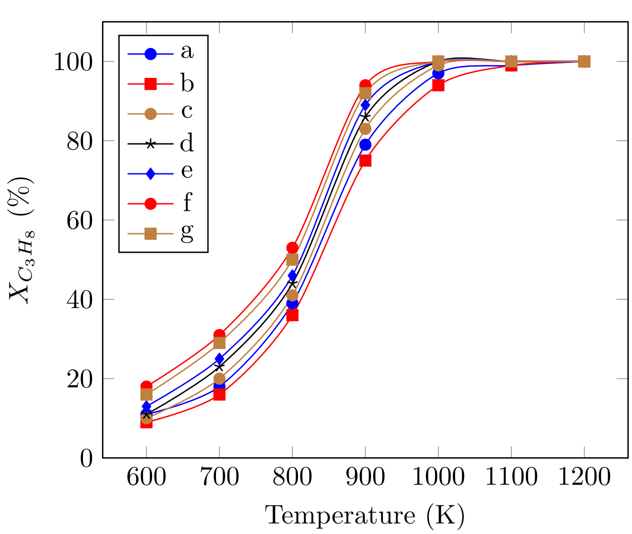

Fig. 2 – Equilibrium conversion rate of propane (C_3H_8) for different feed composition at atmospheric pressure.

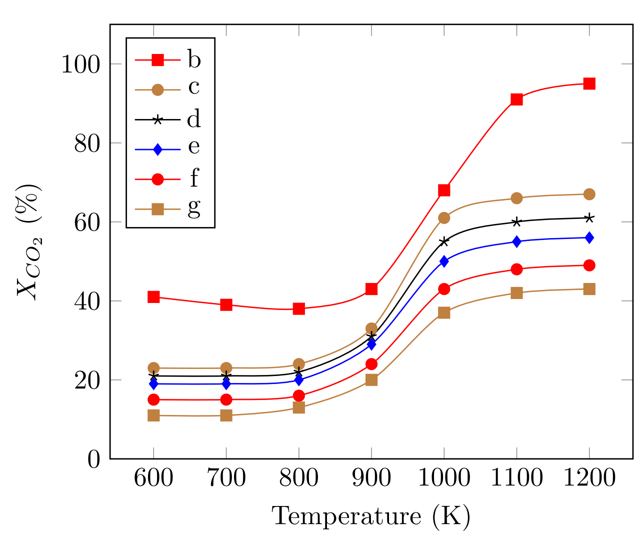

Fig. 4 – Equilibrium conversion rate of carbon dioxide.(CO_2).

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increase numbers of H_2O moles in the feed mixture. For the mixtures high temperature favors CO_2 conversion rate. This can be ascribed to the highly endothermic reaction of propane dry reforming. Besides, over 1000 K, an increasing trend for the rate of CO_2 conversion is observed due to reverse water-gas shift reaction, which is endothermic, favored at high temperatures and consumes carbon dioxide.

CO and H_2 yields

The equilibrium CO and H_2 yields for combined steam-dry propane reforming are shown in Fig. 5 and Fig. 6, respectively.

The influence of H_2O addition is less significant on carbon monoxide yields at all investigated temperature range, but the effect of steam addition on H_2 yield at temperatures above 900 K is observed. For the all feed mixtures, increasing temperature led to increasing both hydrogen and carbon monoxide yields.

For H_2 yield increasing trend dampened above 1000 K and for f-feed composition (Table 3) after 1100 K trend became reversal. These facts happened due to two reasons. First, at high temperatures dry reforming, which yields less hydrogen compared to H_2O propane reforming, and reverse water-gas shift reaction, which contemporaneously consumes hydrogen, are favored. Second, the addition H_2O gave over the fixed value of propane and CO_2 in the inlet gas mixture is represented in the denominator of hydrogen yield equation (Eq. (9)), which also decreased the resulting hydrogen yield.

The synthesis gas (H_2 + CO) production for combined steam-dry propane reforming is shown on Fig. 7. It was observed that the syngas moles increased with increasing temperature to 1100 K at atmospheric pressure. As it shown on Fig. 7, the syngas moles was increased to about 10 mol for the all feed stock (Table 3). At a constant temperature, the syngas moles was increased with increasing number of H_2 moles for temperature until to 1000 K.

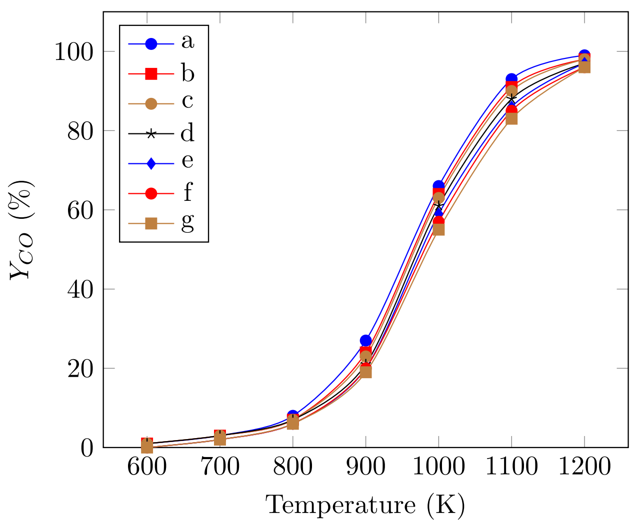

Fig. 5 – Equilibrium CO yield as a function of temperature.

Methane and carbon formation

Coking and methane formation are another factors that needs to be investigated notably at low temperatures for combined steam-dry propane reforming. Fig. 8 shows the number of methane moles produced from combined propane reforming as a function and temperature at pressure p = 1 bar.

In total, the number of methane moles reduce with the increasing temperature and number of oxidant moles in the feed stock. In other words, high temperatures and oxidant moles can inhibit the methane production. Thermodynamically, the methane production can be inhibited almost entirely over 1100 K for all the inlet gas compositions, especially with large number of oxidant moles. The maximum number of methane moles produced for the interesting feed stock (c–g according Table 3) for thermochemical waste-heat recuperation in schematic diagram on Fig. 1 is only approximately 1 mol.

For combined steam-dry propane reforming which used for thermochemical waste-heat recuperation, methane formation is not a desirable product because the coking production at high temperatures can be by methane cracking reaction (according Table 2).

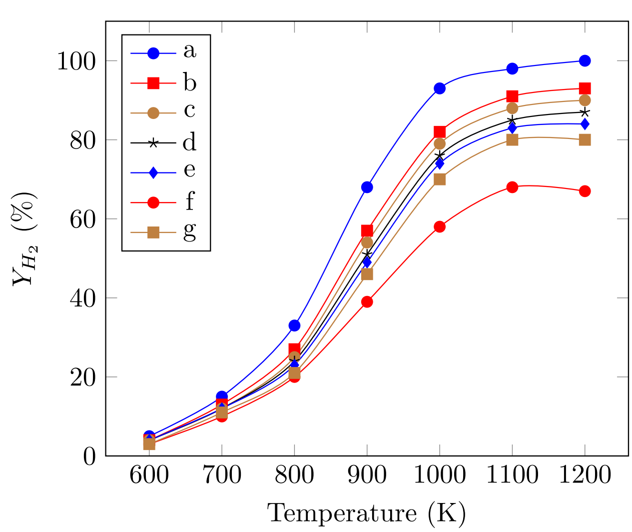

Fig. 6 – Equilibrium H_2 yield as a function of temperature.

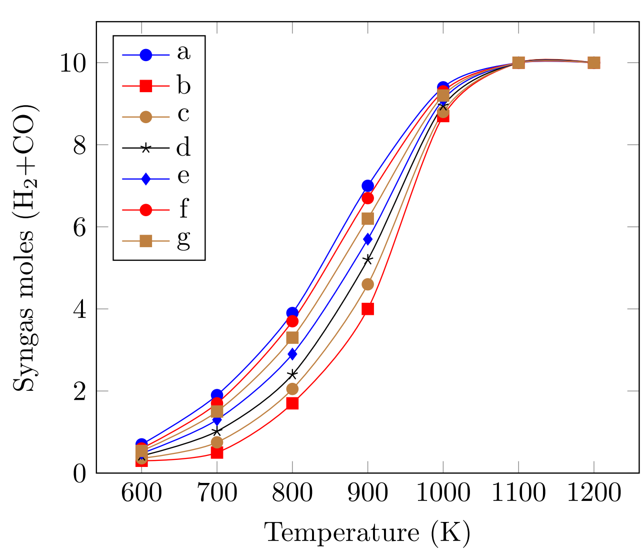

Fig. 7 – Syngas moles in combined DR-SR of propane.

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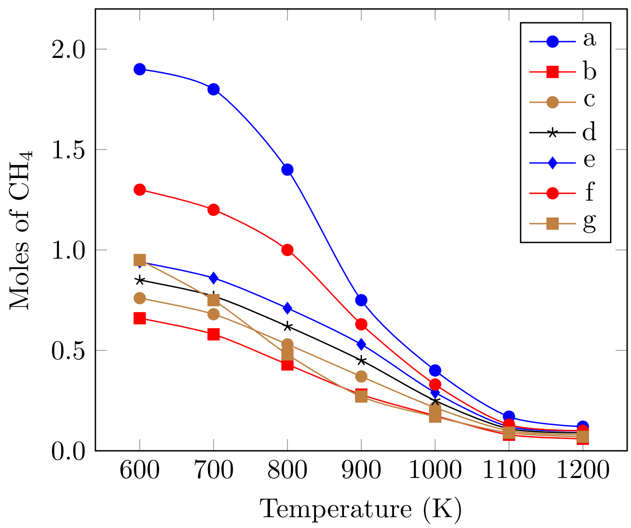

Fig. 8 – Methane (CH_4) moles in combined DR-SR of propane.

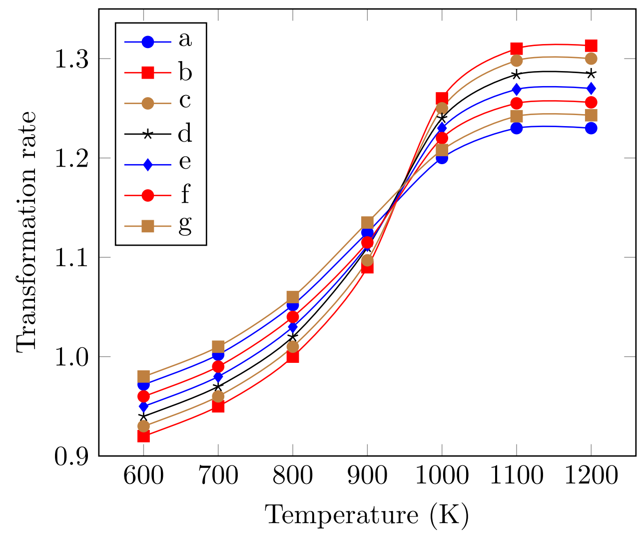

Fig. 10 – The transformation rate as a function of temperature.

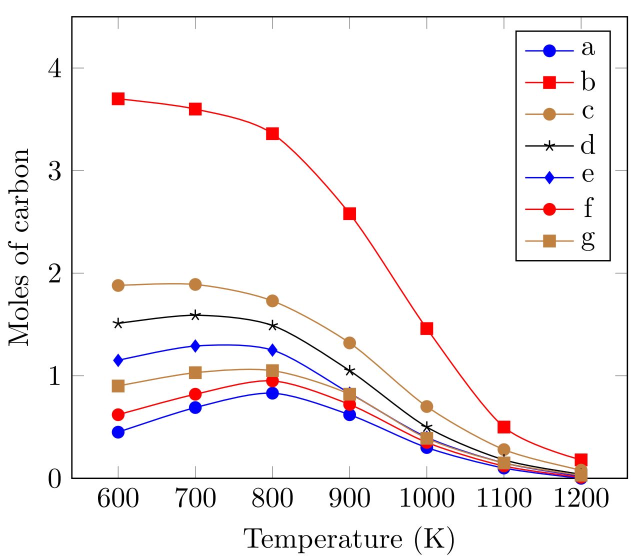

Fig. 9 – Moles of carbon as a function of temperature.

As can be seen from Fig. 9, carbon formation is dependent on temperature for the all feed composition. Moreover, high rate of oxidants to propane favor the inhibition of carbon formation especially at a temperature until 900 K. Carbon formation can also be eliminated under conditions where the maximum amount of synthesis gas can be obtained. Thermodynamically, the carbon formation production can be eliminated almost entirely over 1100 K due to according Fig. 8 methane production at these temperatures is minimum. The problem of carbon formation can be reduced by the presence of addition H_2O in the combined propane reforming. This is why the addition of steam can favorably affect to the reforming process for thermochemical waste-heat recuperation. However, the addition of steam in the feed increases the incombustible gases in the synthetic fuel and thus reduces the efficiency of TCR. The determination of the optimal feed composition for thermochemical waste-heat recuperation by combined propane reforming is an important problem of future studies.

Transformation rate

The transformation rate is the ratio of the low heat value of the product gases in relation to the LHV of C_3H_8. Fig. 10 shows that in increase of reaction temperature increases the LHV of the syngas in comparison to LHV of initial propane.

At temperature below 950 K, the transformation rate decreased with increase the number of oxidant moles at constant temperature while reverse trend was observed for temperature range above 950 K. The transformation rate was about same in between the temperatures about 950 K for the all considered feed stocks. For several considered cases at temperatures below 700–800 K the transformation rate is less unit. At these conditions the thermochemical waste-heat recuperation cannot be carried out. The transformation rate is less than unity because of the reaction enthalpy for overall combined steam-dry propane reforming is positive and the process is exothermic.

Conclusion

The thermodynamic analysis of combined steam-dry propane reforming for thermochemical waste-heat recuperation was investigated via the technique of Gibbs free energy minimization by using software IVTANTHERMO. The effects of temperature and the feed composition on the equilibrium compositions and coke deposition were determined. The using of thermochemical waste-heat recuperation may be considered as one of the steps to hydrogen economy. In the fuel-consuming equipment with TCR the synthesis gas, containing hydrogen, is produced and used as a fuel. In other words, the principle of on-board hydrogen production is performed [3].

The results showed that the using of TCR becomes effective at temperatures above 950 K because in this range the maximum conversion rate is reached (from 1.22 to 1.30 for the different feed composition). Atmospheric pressure is feasible for combined reforming of propane because the compression of the flue gas is energetically unwise. Approximately 10 mol of synthesis gas can be produced over 1000 K from combined steam-dry propane reforming. Furthermore, the propane conversion rate and yield of hydrogen are increased with the addition of extra steam to the feed stock. Also, undesirable carbon formation can be eliminated by adding steam to the feed. However, the addition of steam in the feed increases the non-combustible gases in the synthetic fuel and thus reduces the efficiency of TCR. The determination of the optimal feed composition for thermochemical waste-heat recuperation by combined propane reforming is an important problem of future studies.

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